Order Selection Methods for Autoregressive Processes

Classical, Predictive, Bayesian, and Heavy-Tail Robust Approaches

Generated study note

May 28, 2026

Abstract

This document surveys order selection for univariate autoregressive processes. It starts from the standard finite-variance Gaussian framework, where likelihood, residual variance, partial autocorrelations, and information criteria give a clean and widely used methodology. It then explains why the same tools become delicate under heavy-tailed or infinite-variance innovations, especially for stable innovations with tail index $0 < \alpha < 2$. The main message is that there is no single universally best order selector. AIC-type criteria are natural for prediction, BIC/HQ-type criteria are natural for consistent finite-order identification under regular conditions, cross-validation targets out-of-sample prediction, and robust or distribution-aware criteria are preferable when extremes dominate least-squares fits. For heavy-tailed innovations, one should separate three questions: the dynamic order p, the innovation law, and whether the process is causal, noncausal, or mixed.

Contents

1 The autoregressive order-selection problem **2**

2 Preliminaries: estimation for a fixed order **2**

3 Classical visual and correlation-based methods **2**
3.1 Autocorrelation and partial autocorrelation 2
3.2 Sequential residual tests 3

4 Information criteria **3**
4.1 General form 3
4.2 Akaike information criterion 3
4.3 Bayesian information criterion / Schwarz criterion 3
4.4 Hannan-Quinn criterion 4
4.5 Final prediction error and related criteria 4

5 Cross-validation and rolling forecast evaluation **4**

6 Bayesian order selection **4**

7 Penalized regression and high-dimensional lag selection **5**

8 Residual diagnostics after order selection **5**

9 Heavy-tailed innovations: what changes? **5**
9.1 Classes of heavy-tailed innovations 5
9.2 Why least squares and Gaussian criteria become fragile 6

10 Order selection under heavy tails **6**
10.1 Distribution-aware likelihood criteria 6
10.2 Robust information criteria 6
10.3 Robust cross-validation 7
10.4 Tail-index-aware procedures 7
10.5 Infinite-variance AR model identification 7

11 Practical comparison of methods **7**

12 Recommended workflow **8**
12.1 Step 1: start with exploratory diagnostics 8
12.2 Step 2: choose the target 8
12.3 Step 3: compare several selectors 9
12.4 Step 4: check residuals and stability 9

13 Special remarks on α-stable innovations **9**

14 Connection to geometric or scale-selection viewpoints **9**

15 Summary **10**

1 The autoregressive order-selection problem

A univariate autoregressive process of order p, denoted AR(p), is usually written as

$$X_t = \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + Z_t, \tag{1}$$

where (Z_t) is an innovation sequence. In the classical model, (Z_t) is independent and identically distributed with mean zero and finite variance σ^2. Often one additionally assumes Gaussian innovations, because this turns least squares and maximum likelihood into essentially the same estimation method.

The order-selection problem is: given observations $X_1, \ldots, X_T$, choose a lag order $\widehat{p}$ from a candidate set $\{0, 1, \ldots, p_{\max}\}$. The chosen order can have different intended meanings:

- **Structural order:** the smallest p for which (1) is the true finite-order data-generating mechanism.
- **Predictive order:** the order that gives the best finite-sample forecasts.
- **Approximation order:** the finite order used to approximate an infinite-order or misspecified process.

These goals are related but not identical. AIC is usually closer to predictive or approximation performance, whereas BIC and Hannan-Quinn are usually closer to consistent recovery of a finite true order under regularity conditions.

2 Preliminaries: estimation for a fixed order

For a fixed candidate order m, collect the regressors

$$\mathbf{x}_{t-1}^{(m)} = (X_{t-1}, \ldots, X_{t-m})^\top, \qquad t = m+1, \ldots, T.$$

The conditional least-squares estimator is

$$\widehat{\phi}_m = \operatorname*{arg\,min}_{\phi \in \mathbb{R}^m} \sum_{t=m+1}^{T} \left(X_t - \phi^\top \mathbf{x}_{t-1}^{(m)} \right)^2.$$

The corresponding residuals and residual variance estimate are

$$\widehat{Z}_{t,m} = X_t - \widehat{\phi}_m^\top \mathbf{x}_{t-1}^{(m)}, \qquad \widehat{\sigma}_m^2 = \frac{1}{T-m} \sum_{t=m+1}^{T} \widehat{Z}_{t,m}^2.$$

Under Gaussian innovations, maximizing the conditional likelihood over ϕ is equivalent to minimizing the residual sum of squares. Therefore many order-selection methods can be written as a trade-off between fit, measured by $\widehat{\sigma}_m^2$ or log-likelihood, and complexity, measured by m.

3 Classical visual and correlation-based methods

3.1 Autocorrelation and partial autocorrelation

The autocorrelation function (ACF) and partial autocorrelation function (PACF) are classical diagnostic tools. For an AR(p) process with finite second moments, the PACF cuts off after lag p: the partial autocorrelation is zero for lags larger than p. In practice, one estimates the PACF and looks for a last clearly significant lag.

The advantage is interpretability. The method directly reflects the linear prediction structure of autoregressions: in an AR(p) model, once $X_{t-1}, \ldots, X_{t-p}$ are known, older lags do not improve the best linear predictor. The disadvantage is that finite samples, near-unit roots, seasonality, outliers, and heavy tails can make the visual cutoff unclear.

3.2 Sequential residual tests

Another traditional approach is sequential testing. Fit an AR(m) model and test whether the last coefficient is zero:

$$H_0 : \phi_m = 0 \quad \text{versus} \quad H_1 : \phi_m \neq 0.$$

One can test increasing orders until no further lag appears significant, or decreasing orders from $p_{\max}$ until the last coefficient is significant. Residual whiteness tests, such as portmanteau tests, are then used to check whether remaining serial correlation is present.

Sequential tests are intuitive, but they depend on significance levels, repeated-testing conventions, and asymptotic approximations. They are therefore less clean as automatic order selectors than information criteria.

4 Information criteria

4.1 General form

Information criteria choose the order minimizing

$$IC(m) = -2\ell_m + \text{penalty}(m, T), \tag{2}$$

where ℓ_m is the maximized log-likelihood under order m. In the conditional Gaussian AR case, this is equivalent up to constants to criteria of the form

$$IC(m) = T \log(\widehat{\sigma}_m^2) + c_T m. \tag{3}$$

Here c_T controls the strength of the complexity penalty. A larger penalty tends to choose smaller models.

4.2 Akaike information criterion

AIC uses $c_T = 2$:

$$AIC(m) = T \log(\widehat{\sigma}_m^2) + 2m. \tag{4}$$

Akaike introduced AIC as an information-theoretic model identification criterion, motivated by estimating expected Kullback-Leibler discrepancy rather than by selecting the true finite-dimensional model with probability tending to one [1]. For autoregressive approximation, AIC is often good for forecasting because it is willing to keep additional lags that reduce prediction error.

AIC tends to overselect in finite true-order settings: even when the true process is exactly AR(p), AIC may choose orders larger than p with non-vanishing probability asymptotically. This is not necessarily a defect if the target is prediction rather than exact structural recovery.

4.3 Bayesian information criterion / Schwarz criterion

BIC, also called the Schwarz criterion, uses $c_T = \log T$:

$$BIC(m) = T \log(\widehat{\sigma}_m^2) + m \log T. \tag{5}$$

Schwarz derived the criterion from an asymptotic Bayesian approximation to integrated likelihoods [2]. Since $\log T$ grows, BIC penalizes complexity more strongly than AIC in large samples.

Under standard regularity conditions and if the true process has a finite AR order contained in the candidate set, BIC is usually consistent: the selected order converges in probability to the true order. However, because of its stronger penalty, BIC can underfit in small samples or when the true process is better viewed as a high-order approximation problem.

4.4 Hannan-Quinn criterion

The Hannan-Quinn criterion uses the intermediate penalty

$$HQ(m) = T \log(\widehat{\sigma}_m^2) + 2m \log \log T. \tag{6}$$

Hannan and Quinn introduced it specifically for determining the order of an autoregression [3]. Its penalty grows more slowly than BIC but faster than AIC. The criterion is important theoretically because the $\log \log T$ rate is close to the minimal growth rate needed for strong consistency in many autoregressive order-selection problems.

4.5 Final prediction error and related criteria

Akaike's final prediction error (FPE) criterion estimates out-of-sample one-step prediction error. For an AR(m) model it is often written as

$$FPE(m) = \widehat{\sigma}_m^2 \frac{T+m}{T-m}. \tag{7}$$

For moderate or large T, minimizing FPE is closely related to minimizing AIC. The intuitive idea is simple: adding lags always improves in-sample residual variance, so the criterion inflates the residual variance by a factor that accounts for estimation uncertainty.

5 Cross-validation and rolling forecast evaluation

Cross-validation selects p by measuring prediction error on data not used to fit the model. For time series, random train-test splits are inappropriate because they break temporal dependence. Instead one usually uses rolling-origin evaluation:

1. Choose candidate orders $m = 0, \ldots, p_{\max}$.
2. For each forecast origin s, fit the AR(m) model using observations up to time s.
3. Predict X_{s+1} and record a loss $L(X_{s+1}, \widehat{X}_{s+1|s}^{(m)})$.
4. Average the loss over forecast origins and select the order with smallest average loss.

The common squared loss is

$$L_2(e) = e^2,$$

but for heavy tails it can be more stable to use absolute loss $L_1(e) = |e|$, Huber loss, quantile loss, or trimmed squared loss.

Cross-validation is especially useful when the model is treated as a forecasting device. Its weakness is computational cost and dependence on the chosen validation scheme, horizon, and loss function.

6 Bayesian order selection

Bayesian order selection treats the order as an unknown model index. One specifies prior probabilities $\Pr(p = m)$ and parameter priors $\pi_m(\phi_m, \sigma^2)$, then computes posterior model probabilities

$$\Pr(p = m \mid X_{1:T}) \propto p(X_{1:T} \mid p = m) \Pr(p = m),$$

where

$$p(X_{1:T} \mid p = m) = \int p(X_{1:T} \mid \phi_m, \sigma^2, p = m) \pi_m(\phi_m, \sigma^2) \, d\phi_m \, d\sigma^2$$

is the marginal likelihood. The selected order can be the posterior mode, or one can average forecasts over orders.

BIC can be interpreted as a large-sample approximation to this Bayesian comparison. Full Bayesian selection is more flexible: it can incorporate shrinkage priors, heavy-tailed likelihoods, prior beliefs about low order, and uncertainty in p. The price is computational complexity and sensitivity to priors, especially in small samples.

7 Penalized regression and high-dimensional lag selection

Instead of choosing one order from nested models, one may fit a large autoregression and penalize coefficients:

$$\widehat{\phi} = \arg\min_{\phi} \sum_{t} \left(X_t - \sum_{j=1}^{p_{\max}} \phi_j X_{t-j} \right)^2 + \lambda \sum_{j=1}^{p_{\max}} |\phi_j|.$$

The lasso can set some lag coefficients exactly to zero. If one wants an actual AR order, one can define

$$\widehat{p} = \max\{j : \widehat{\phi}_j \neq 0\}.$$

This approach is useful when $p_{\max}$ is large, when sparsity is plausible, or in multivariate VAR settings. However, lasso-type methods select active lags, not necessarily a contiguous order. For AR order selection, one may prefer hierarchical lag penalties that enforce the principle: if lag j is active, lower lags should also be included.

8 Residual diagnostics after order selection

No order selector should be used without diagnostic checks. After selecting $\widehat{p}$, examine:

- residual ACF/PACF;
- portmanteau tests for remaining autocorrelation;
- residual distribution and tail behavior;
- stability of the fitted AR polynomial;
- forecast performance on held-out periods;
- sensitivity of $\widehat{p}$ to $p_{\max}$ and to individual extreme observations.

For heavy-tailed data, residual diagnostics should not rely only on variance-based summaries. Quantile plots, empirical characteristic functions, tail-index estimates, and robust residual losses are often more informative.

9 Heavy-tailed innovations: what changes?

9.1 Classes of heavy-tailed innovations

Heavy-tailed innovations appear in finance, network traffic, hydrology, insurance, turbulence, and other settings with bursts or extreme shocks. Common model classes include:

- Student-t innovations, with finite variance only when degrees of freedom exceed 2;
- generalized error and Laplace-type innovations;
- scale mixtures of normals;

- Pareto-like innovations;
- symmetric or skewed α-stable innovations.

An α-stable innovation has tail index $0 < \alpha \leq 2$. The Gaussian distribution is the special stable case $\alpha = 2$. For $\alpha < 2$, the variance is infinite. For $\alpha \leq 1$, even the mean is not finite. This matters because many classical AR tools are built around second moments, residual variance, and Gaussian likelihood.

9.2 Why least squares and Gaussian criteria become fragile

Least squares minimizes squared residuals. A single very large innovation can dominate the objective, distort coefficient estimates, and strongly affect the residual variance. In an autoregression, an extreme innovation also propagates into lagged regressors, so outliers can become leverage points in subsequent observations.

Consequently, for heavy-tailed innovations:

- sample autocorrelations can be unstable or theoretically undefined when variance is infinite;
- PACF cutoffs become less reliable;
- Gaussian log-likelihood is misspecified;
- AIC/BIC based on $\log(\widehat{\sigma}_m^2)$ may be driven by extremes;
- asymptotic reference distributions for tests may fail or converge slowly.

This does not mean classical criteria are useless. It means that their interpretation changes. They may still be good quasi-likelihood or prediction devices, but they are no longer automatically justified by finite-variance Gaussian theory.

10 Order selection under heavy tails

10.1 Distribution-aware likelihood criteria

One direct response is to replace the Gaussian likelihood by a likelihood that matches the innovation distribution. For Student-t innovations, choose m by minimizing

$$IC_t(m) = -2\ell_{t,m} + c_T k_m,$$

where $\ell_{t,m}$ is the maximized Student-t log-likelihood and k_m includes AR coefficients, scale, and tail parameters. This approach is robust relative to Gaussian likelihood because large residuals receive less extreme quadratic punishment.

For stable innovations, likelihood evaluation is harder because stable densities generally do not have closed forms. Nevertheless, stable autoregressive models can be estimated by numerical likelihood, characteristic-function methods, quantile methods, or simulation-based methods. Work on infinite-variance autoregressions shows that model identification remains possible in such settings, but the theory and algorithms differ from the Gaussian case [6].

10.2 Robust information criteria

A second response is to keep the AR regression structure but replace the squared residual loss by a robust loss. Let ρ be Huber's loss, Tukey's bisquare loss, absolute loss, or a trimmed loss. For each order m, estimate

$$\widehat{\phi}_m^{\rho} = \arg\min_{\phi} \sum_t \rho\left(\frac{X_t - \phi^\top \mathbf{x}_{t-1}^{(m)}}{s_m}\right),$$

where s_m is a robust scale estimate. Then choose m by minimizing a robust criterion of the schematic form

$$RIC(m) = \sum_t \rho\left(\frac{\widehat{Z}_{t,m}}{\widehat{s}_m}\right) + \lambda_T \, df_m.$$

Robust model-choice ideas go back at least to robust versions of AIC and robust penalized criteria; Ronchetti discusses robustness aspects of model choice including a robust AIC application to autoregressive models [7], and robust penalized model selection has also been developed in regression contexts [8].

The key design choices are the robust loss, robust scale estimate, and penalty. A robust criterion can be much more stable than Gaussian AIC/BIC when a few extreme innovations dominate the sample.

10.3 Robust cross-validation

For forecasting, robust cross-validation is often the most transparent method. Instead of measuring squared forecast error, use a loss such as

$$L(e) = |e|, \qquad L(e) = \min(e^2, c^2), \qquad L(e) = \rho_c(e).$$

Then choose the order with the best rolling-origin robust validation loss. This avoids relying on a finite variance and directly targets the practical forecasting objective.

A useful compromise is to compute several losses simultaneously: squared loss for ordinary mean-square prediction, absolute loss for median-like robustness, and tail-weighted or quantile loss if extreme risk is important.

10.4 Tail-index-aware procedures

For stable or Pareto-like data, it can be useful to estimate the tail index α and adapt the order-selection method accordingly. If $\widehat{\alpha}$ is near 2, classical finite-variance tools may be acceptable. If $\widehat{\alpha} < 2$, variance-based criteria should be treated cautiously. Mudelsee and Bermejo propose data-adaptive order selection for heavy-tail estimation and study simulations generated from stable distributions with AR(1) serial dependence [9]. Although their target is tail-estimation order rather than AR lag order, the paper is relevant because it shows how serial dependence and stable heavy tails interact with order-type tuning decisions.

10.5 Infinite-variance AR model identification

A particularly relevant result is Andrews' work on model identification for infinite-variance autoregressive processes. The paper considers infinite-variance AR time series and shows that a consistent estimate of the autoregressive model order can be obtained by minimizing AIC in that setting [6]. This is striking because, in the standard finite-variance fixed-order Gaussian intuition, AIC is usually not the consistent structural order selector. The point is not that ordinary AIC is always safe under heavy tails; rather, the asymptotic behavior can change substantially when the innovation distribution has infinite variance.

This literature is also connected to noncausal and mixed autoregressions. With non-Gaussian heavy-tailed innovations, the distinction between causal and noncausal roots can become identifiable from the distributional shape, whereas in Gaussian linear models causal and noncausal representations can be observationally equivalent in second-order terms.

11 Practical comparison of methods

Method	Main target	Strengths	Weaknesses
PACF cutoff	Structural AR order under finite variance	Simple, interpretable, fast	Visual, unstable under heavy tails/outliers
AIC/FPE	Prediction / KL approximation	Good forecasting behavior, easy to compute	May overselect; Gaussian version fragile under extremes
BIC	Consistent finite-order recovery	Strong asymptotic justification for true finite order	Can underfit; sensitive to misspecification
HQ	Strongly consistent order selection	Intermediate penalty; classical for AR order	Less common in software/practice
Sequential tests	Hypothesis-driven lag inclusion	Transparent inferential logic	Multiple testing; asymptotic tests can fail under heavy tails
Rolling CV	Forecasting	Directly evaluates predictive target	Computationally heavier; depends on loss/split
Bayesian selection	Posterior model uncertainty	Can average over orders; flexible likelihoods	Prior and computational sensitivity
Robust IC	Heavy tails/outliers	Less dominated by extremes	Requires robust loss, scale, and penalty calibration
Stable/t likelihood IC	Distribution-aware order	Models heavy tails directly	More complex likelihood and optimization
Penalized lag selection	Sparse high-order dynamics	Useful for large $p_{\max}$ and VARs	Active lag set may not equal contiguous order

12 Recommended workflow

A practical workflow for AR order selection is as follows.

12.1 Step 1: start with exploratory diagnostics

Plot the series, inspect obvious extremes or regime changes, and compute ACF/PACF as descriptive tools. If the series contains extreme bursts, avoid relying solely on variance-based plots.

12.2 Step 2: choose the target

If the goal is structural identification of a finite-order AR model, compare BIC and HQ. If the goal is forecasting, compare AIC, FPE, and rolling cross-validation. If the goal is a robust model in the presence of extremes, use robust cross-validation or robust information criteria.

12.3 Step 3: compare several selectors

Compute, for $m = 0, \ldots, p_{\max}$,

$$AIC(m), \quad BIC(m), \quad HQ(m), \quad CV_{L_1}(m), \quad CV_{L_2}(m),$$

and, when heavy tails are suspected,

$$CV_{Huber}(m), \quad RIC(m), \quad IC_t(m).$$

Agreement across methods is strong evidence. Disagreement is informative: for example, AIC choosing a higher order than BIC often indicates that extra lags improve prediction but are not strongly supported as structural parameters.

12.4 Step 4: check residuals and stability

After selecting $\widehat{p}$, inspect residual dependence and tail behavior. If residuals remain autocorrelated, increase $p_{\max}$ or consider ARMA/GARCH extensions. If residuals remain heavy-tailed, prefer a heavy-tailed likelihood or robust inference for uncertainty quantification.

13 Special remarks on α-stable innovations

For α-stable innovations with $\alpha < 2$, the variance is infinite. Therefore, methods based on minimizing estimated variance should be interpreted as quasi-methods rather than literal variance estimators. Several adaptations are natural:

- use LAD or quantile autoregression instead of least squares;
- use empirical characteristic functions or codifference instead of autocovariance;
- use stable likelihood or approximate stable likelihood when feasible;
- use robust rolling validation losses;
- estimate the tail index and report sensitivity of $\widehat{p}$ to tail modeling choices.

For symmetric α-stable ARMA models, codifference and normalized codifference provide dependence measures that remain meaningful when covariance is unavailable; Rosadi studies estimation of codifference for ARMA processes with symmetric α-stable innovations [10]. This is relevant for order selection because classical covariance-based identification is precisely what breaks down in infinite-variance settings.

14 Connection to geometric or scale-selection viewpoints

If one embeds a scalar time series by sliding windows,

$$Y_t^{(k)} = (X_t, X_{t-1}, \ldots, X_{t-k+1}) \in \mathbb{R}^k,$$

then the AR order becomes geometrically visible in the dependence structure of the embedded point cloud. For an AR(p) process, the coordinate X_t is linearly predicted from the previous p coordinates plus an innovation. Thus, when $k = p + 1$, the embedded cloud is concentrated around a hyperplane in $\mathbb{R}^{p+1}$ under light-tailed innovations. Under bounded innovations, it may occupy a slab or zonotope-like region; under heavy-tailed innovations, the same central linear geometry can be present but with long radial excursions.

This gives a possible bridge between classical order selection and scale-selection methods. Instead of asking only which p minimizes an information criterion, one can examine how geometric or topological summaries of the delay embedding change with k. Heavy tails make this more interesting: extremes may create spurious large-scale geometry, so robust filtrations, trimmed point clouds, rank-based distances, or tail-aware weighting may be needed before interpreting a selected embedding dimension as an AR order.

15 Summary

For ordinary finite-variance AR models, the standard toolkit is well established: PACF for intuition, AIC/FPE for prediction, BIC/HQ for consistent finite-order selection, and residual diagnostics for validation. For heavy-tailed innovations, especially α-stable innovations, the key issue is that variance and covariance may be undefined or unstable. Then one should either model the heavy-tailed likelihood directly, use robust losses and validation criteria, or use dependence summaries that do not require second moments. The safest empirical practice is to report a small panel of order estimates and explain which target each one optimizes.

References

[1] H. Akaike. "A New Look at the Statistical Model Identification." *IEEE Transactions on Automatic Control*, 19(6), 716–723, 1974.

[2] G. Schwarz. "Estimating the Dimension of a Model." *The Annals of Statistics*, 6(2), 461–464, 1978.

[3] E. J. Hannan and B. G. Quinn. "The Determination of the Order of an Autoregression." *Journal of the Royal Statistical Society: Series B*, 41(2), 190–195, 1979.

[4] R. Shibata. "Selection of the Order of an Autoregressive Model by Akaike's Information Criterion." *Biometrika*, 63(1), 117–126, 1976.

[5] P. J. Brockwell and R. A. Davis. *Introduction to Time Series and Forecasting.* 3rd edition, Springer, 2016.

[6] B. Andrews, M. Calder, and R. A. Davis. "Model Identification for Infinite Variance Autoregressive Processes." *Journal of Econometrics*, 172(2), 222–234, 2013.

[7] E. Ronchetti. "Robustness Aspects of Model Choice." *Statistica Sinica*, 7, 327–338, 1997.

[8] S. Mueller and A. Welsh. "Outlier Robust Model Selection in Linear Regression." *Journal of the American Statistical Association*, 100(472), 1297–1310, 2005.

[9] M. Mudelsee and M. Bermejo. "Optimal Heavy Tail Estimation - Part 1: Order Selection." *Nonlinear Processes in Geophysics*, 24, 685–695, 2017.

[10] D. Rosadi. "Estimating the Codifference Function of Linear Time Series Models with Infinite Variance." *Metrika*, 73, 395–429, 2011.

[11] J.-M. Dufour and J. Taamouti. "Point-Optimal Instruments and Generalized Anderson-Rubin Procedures for Nonlinear Models." Related robust inference themes for heavy-tailed dependent data.